ICT305 Topics in IT

Topic 2
Internet of Things (IoT)

CIHE Australia

Agenda

1. Motivation and history of Internet of Things (IoT)
2. Definition of IoT and how does IoT work?
3. Example of an IoT device and application
4. Importance of IoT
5. IoT Applications
6. Challenges of IoT
7. Future for IoT
8. References

History of Internet of Things (IoT)

- In1999, the term "Internet of Things" was coined by Kevin Ashton during his work at Procter &Gamble which became widely accepted. He proposed to use RFID chips on products for supply chain management.
- In 2004, the term was mentioned in popular publications like the Guardian, Boston Globe, and Scientific American.
- In 2005, UN's International Telecommunications Union (ITU) published its first report on this topic.
- In 2008, the Internet of Things (IoT) was born.
- In 2011, Gartner, a market research company, firstly included "The Internet of Things" technology in their research

Motivation of IoT

- The internet of things seeks to eliminate the need for human intervention in several aspects, while making life easier for those who use it.
- **The increase in the number of users** (corporations, companies, SMEs, individuals) who use IoT can be explained by the many benefits it offers such as:

  (1) improving the customer experience;

  (2) improving efficiency;

  (3) access to data/collection of new data;

  (4) reduction of labor cost/actual costs;

  (5) connectivity

Definition of IoT

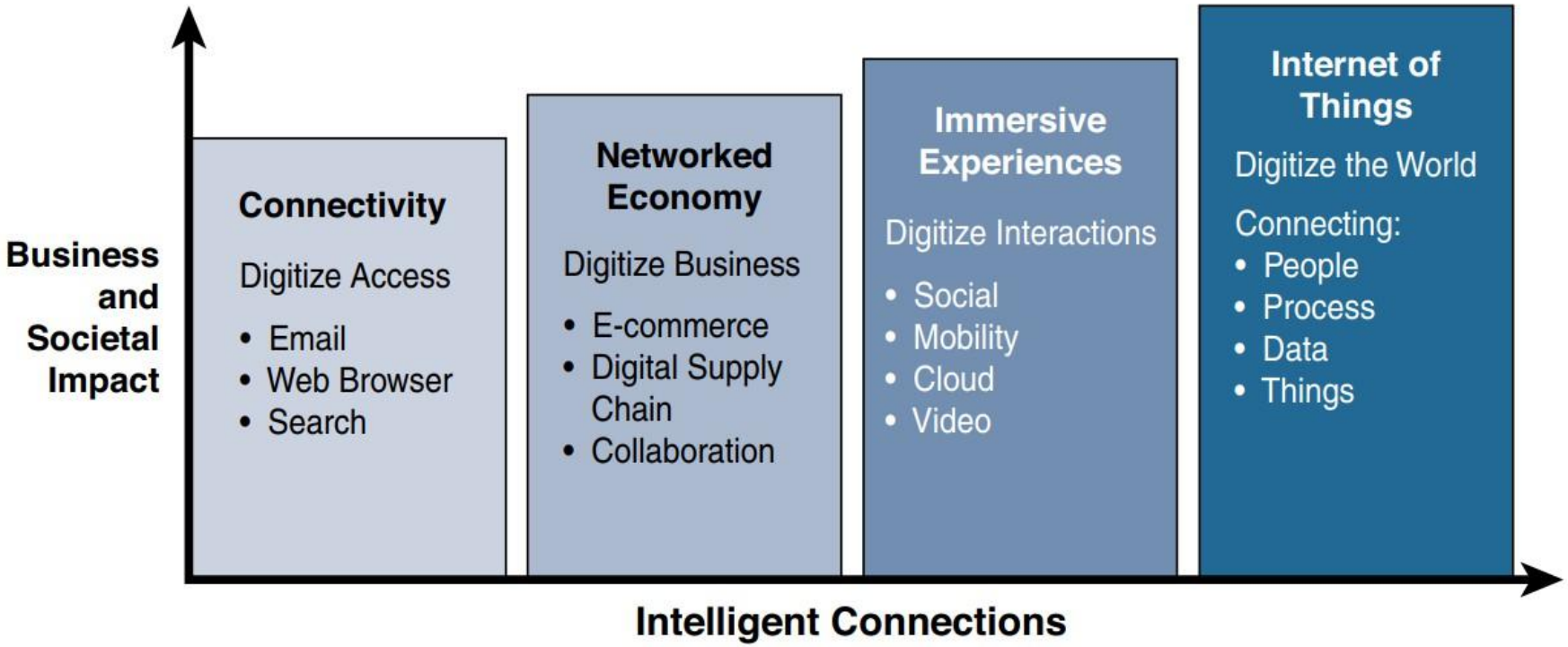
- Imagine all the things in your house are connected.
- They can *communicate* with each other.
- They can *read your mind* and do what you want by reading the *circumstances* around you.
- You can monitor your grandmother's health condition from your class *far away* from her.
- You can switch on/off the fan, lights, window of your bedroom from your office.

IoT Characteristics

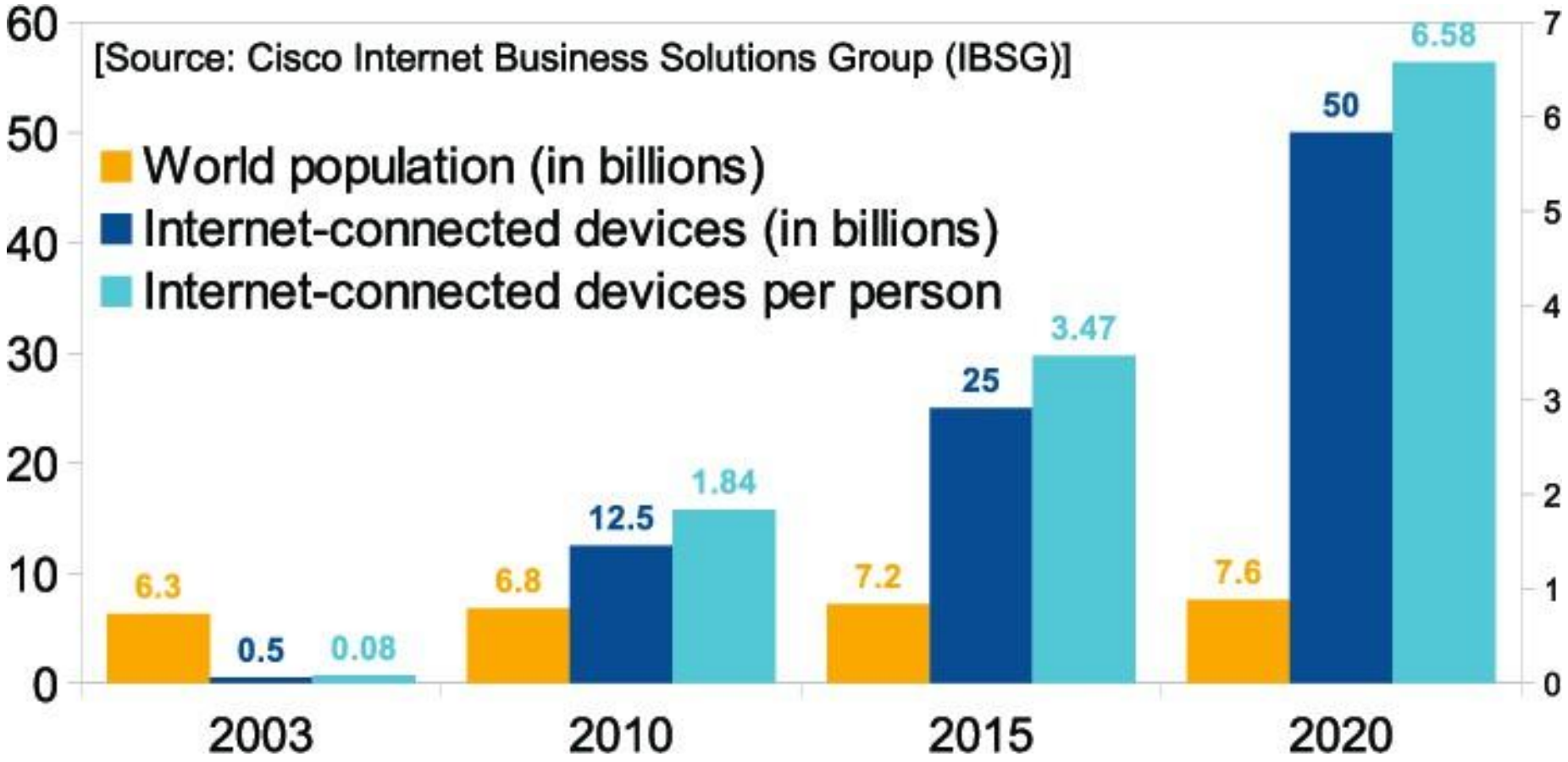
- The things should be:
  - Distinguishable and have unique identification
  - Able to detect the presence of other objects
  - Able to capture data autonomously.
  - Interoperable among various communication technologies
  - Have a secure and fail-safe operation
  - Operate at low power.
  - Programmable by the user
  - Contextual in nature

Evolution of IoT

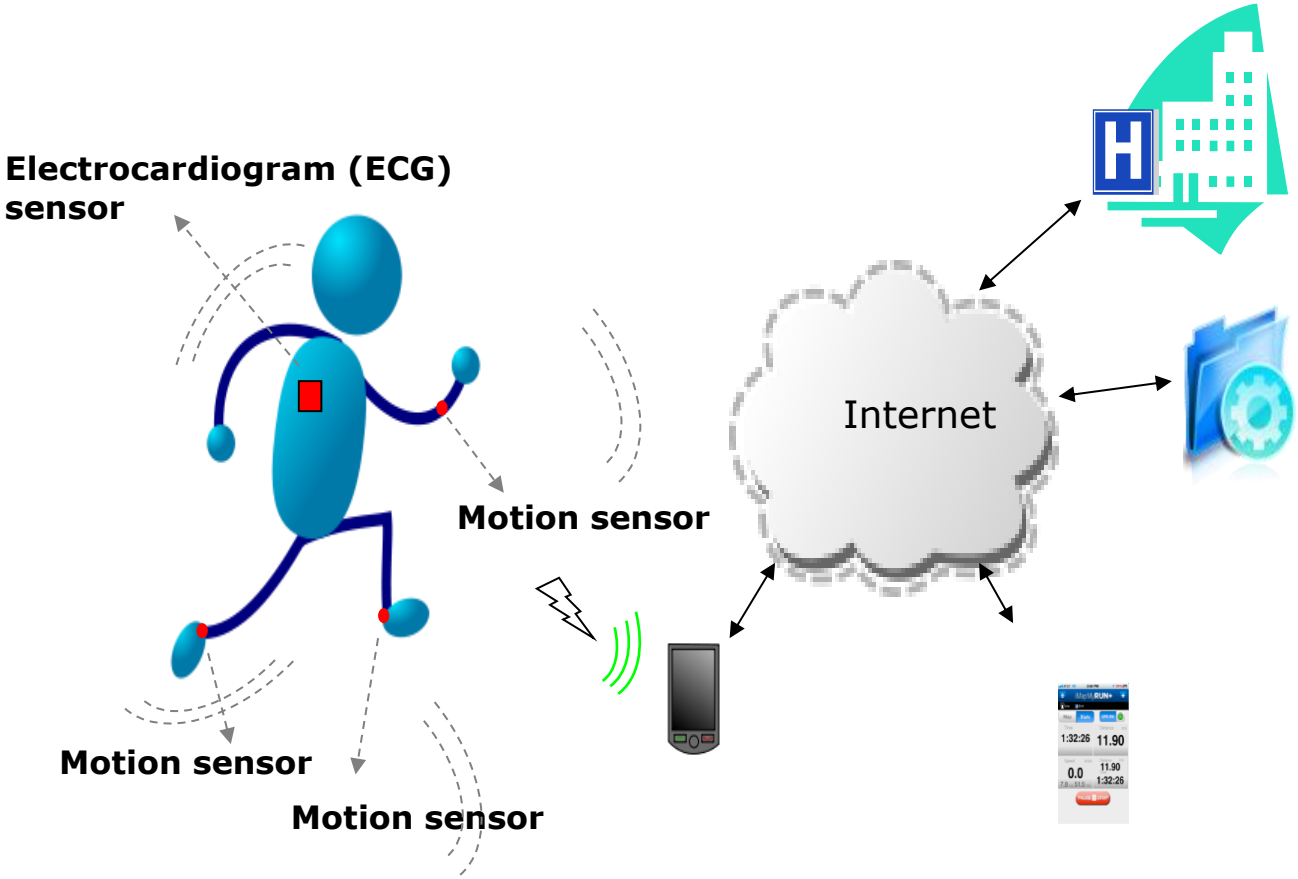
- IoT started at the middle of 2008 and 2009. Its evolution depends on Internet evolution.



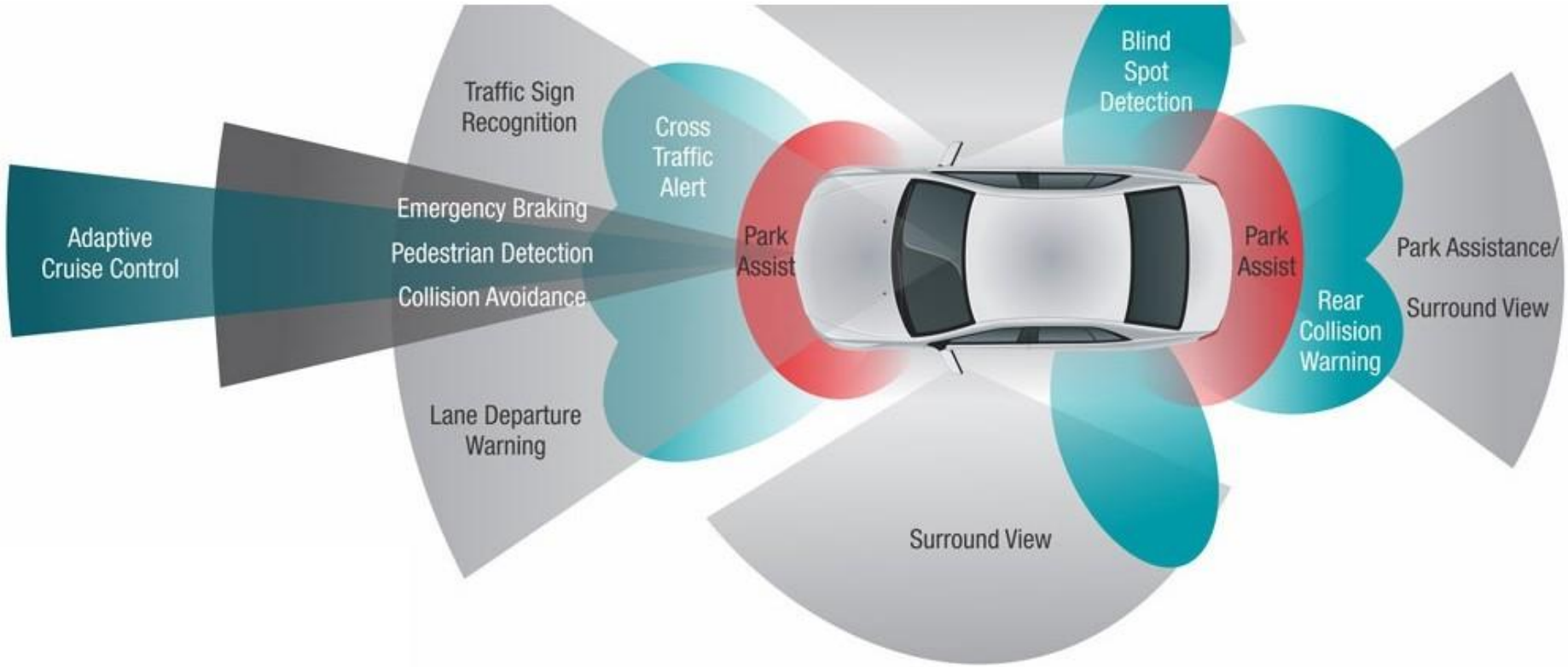
Impact of IoT



IoT: People connecting with Things

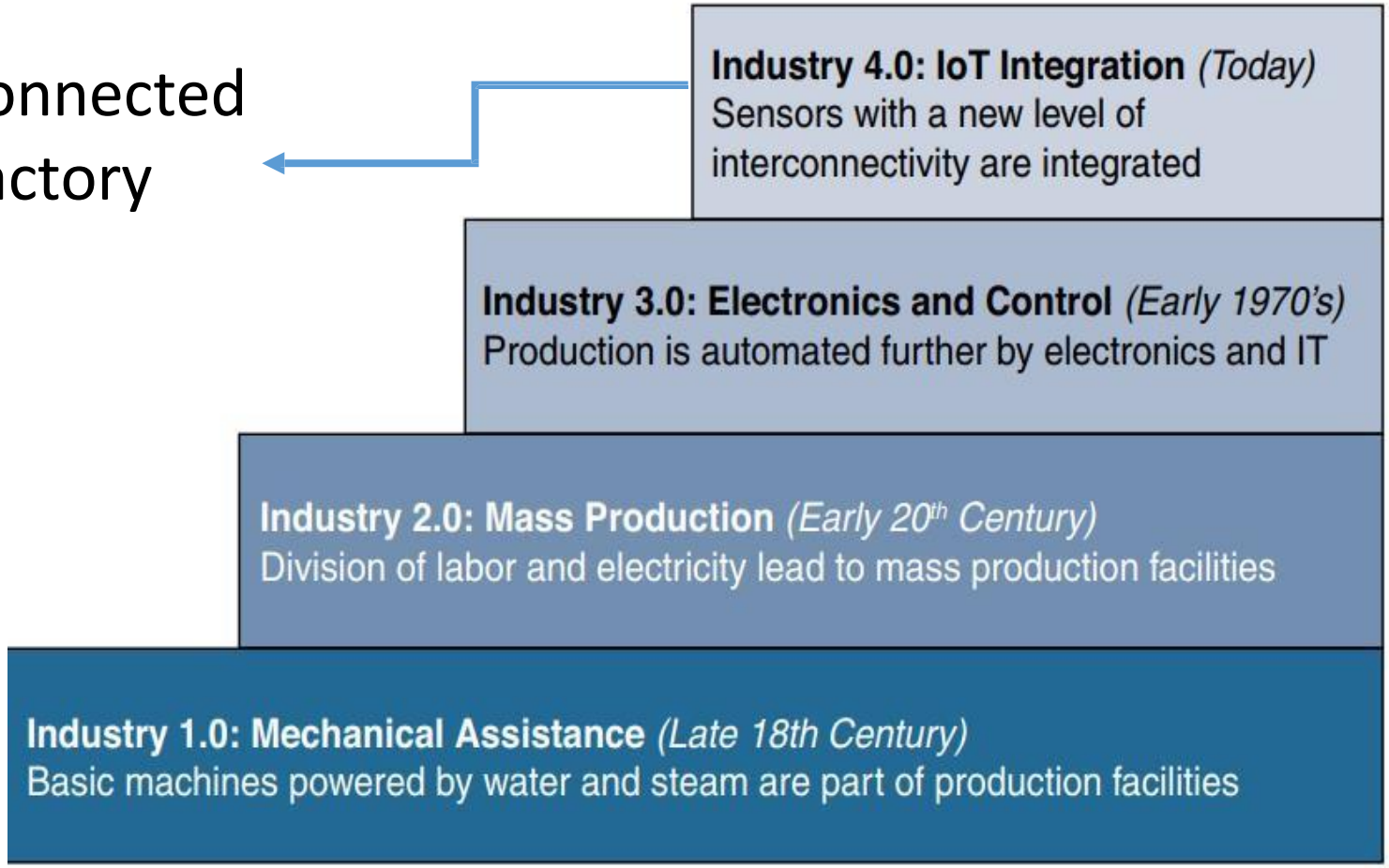


Impact of IoT: Connected Car



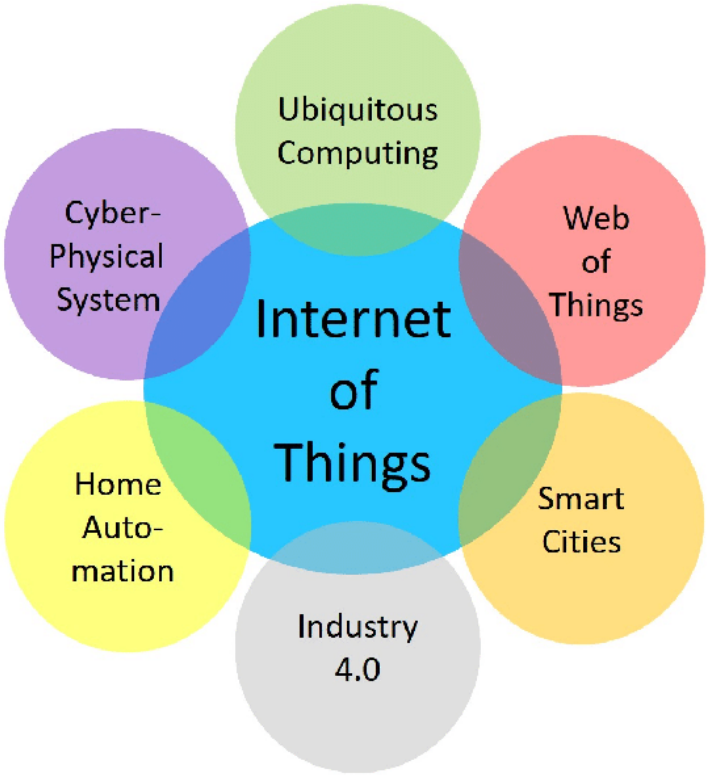
Impact of IoT: 4th Industrial Revolution

Connected Factory



Related concepts to IoT

- M2M (Machine to machine) communication
- Industrial internet (of Things)
- Web of Things
- Industry 4.0
- Smart systems
- Pervasive computing/Ubiquitous Computing
- Intelligent systems



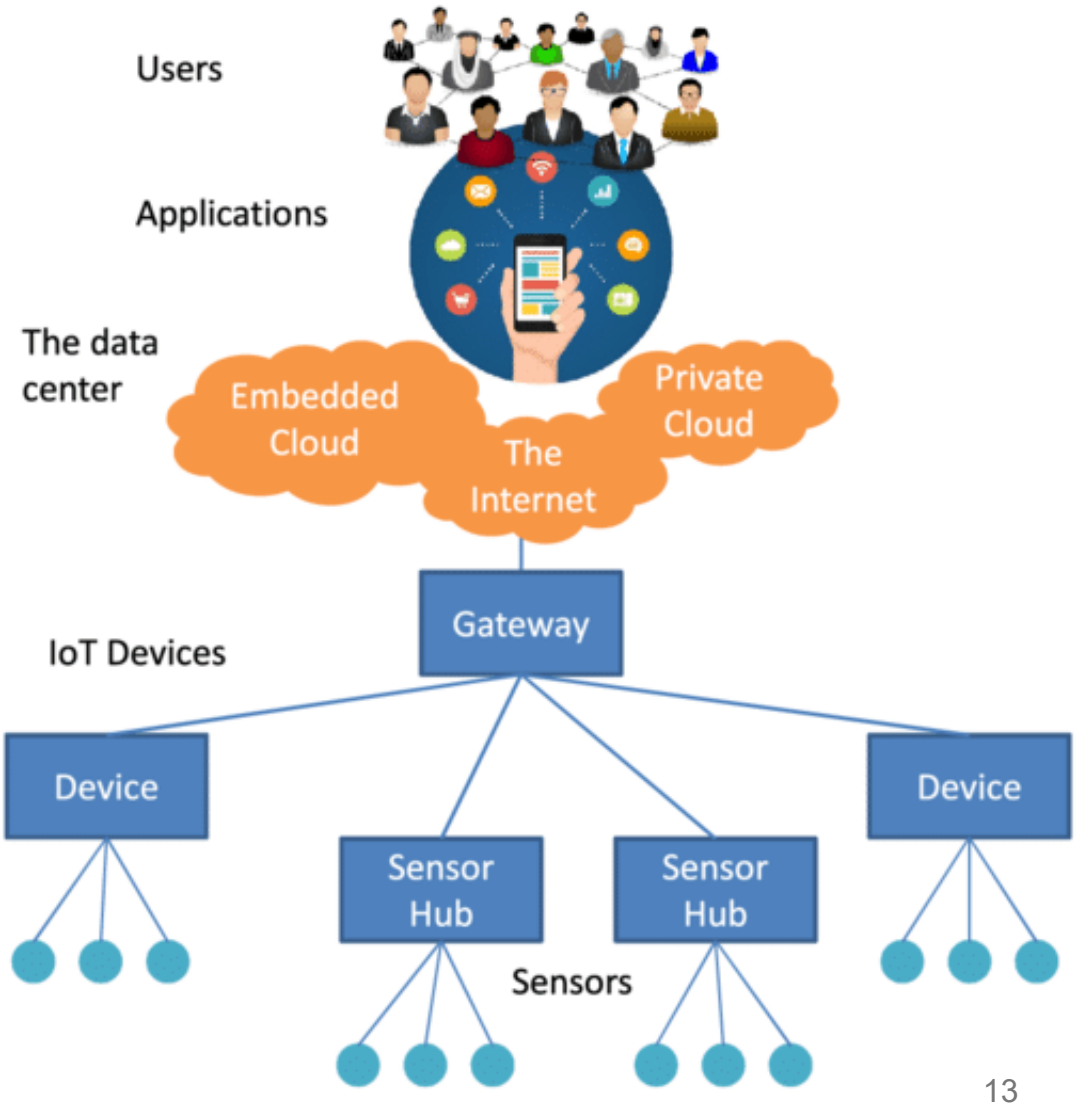
How does IoT work?

1) Sensors/Devices

2) Connectivity

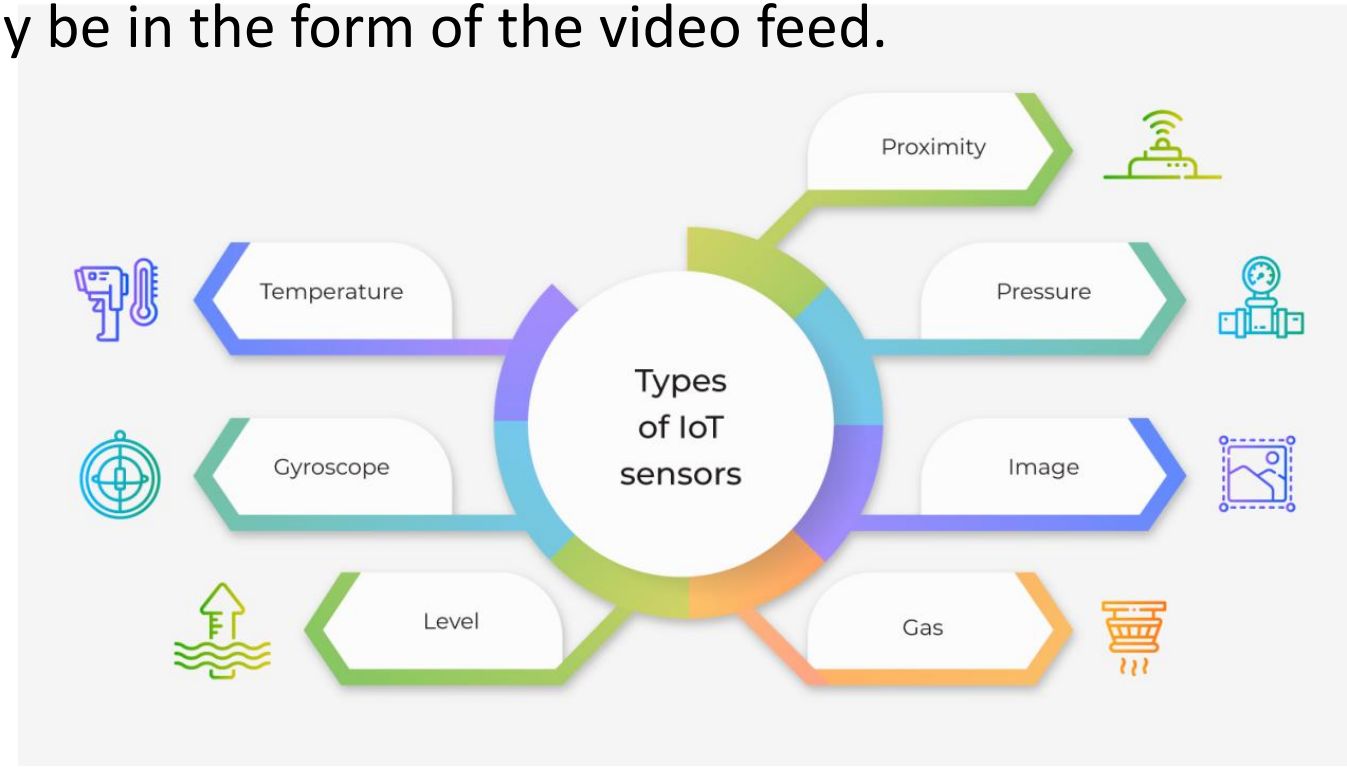
3) Data Processing

4) User Interface



Sensors/Devices

- Sensors or devices are a key component that helps you to collect live data from the surrounding environment.
- All this data may have various levels of complexities.
- It could be a simple temperature monitoring sensor, or it may be in the form of the video feed.



Connectivity

- All the collected data is sent to a cloud infrastructure.
- The sensors should be connected to the cloud using various mediums of communications.
- These communication mediums include mobile or satellite networks, Bluetooth, WI-FI, WAN, etc.

Data Processing

- Once that data is collected, and it gets to the cloud, the software performs processing on the gathered data.
- This process can be just checking the temperature, reading on devices like AC or heaters.
- However, it can sometimes also be very complex like identifying objects, using computer vision on video.

User Interface

- The information needs to be available to the end-user in some way which can be achieved by triggering alarms on their phones or sending them notification through email or text message.
- The user sometimes might need an interface which actively checks their IoT system.
- For example, the user has a camera installed in his home. He wants to access video recording and all the feeds with the help of a web server.

How does IoT work?

A popular example from Youtube advertisement:

Google Home Security

- The IoT is the key driver behind a completely smart and secure home.
- IoT connects a variety of sensors, alarms, cameras, lights, and microphones to provide 24/7/365 security—all of which can be controlled from a smart phone.
- For example, the ring doorbell camera security system allows users to see, hear, and speak to visitors at their door via a computer, tablet, or mobile phone.

Examples of IoT

- A **lightbulb** that can be switched on using a smartphone app is an IoT device, as is a **motion sensor** or a **smart thermostat** in your office or a connected streetlight.
- Bulb control through a mobile app
- Some **larger objects** may themselves be filled with many smaller IoT components, such as a jet engine that's now filled with thousands of sensors collecting and transmitting data back to make sure it is operating efficiently.

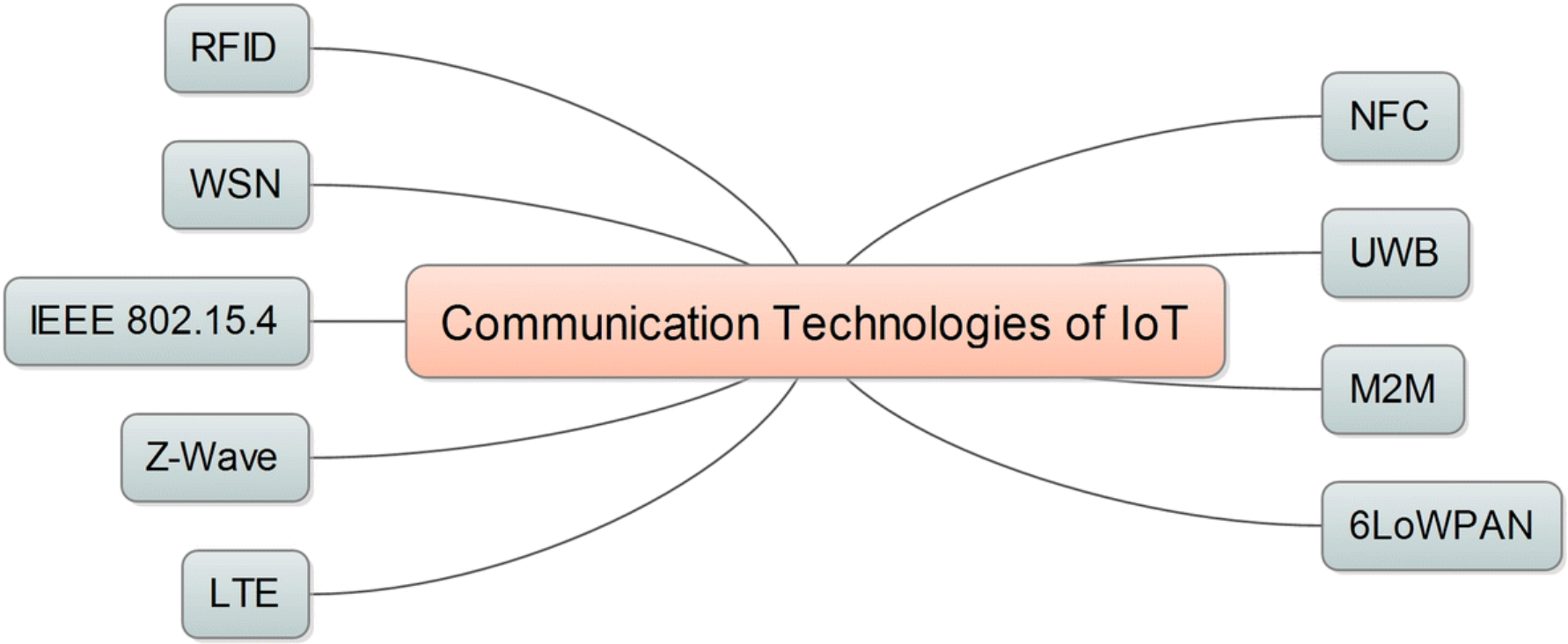
Benefits of IoT to business

- Monitor their overall business processes to improve productivity
- Improve the customer experience (CX)
- Make better business decisions to generate more revenue.

Importance of IoT

- Over the past few years, IoT has become one of the most important technologies of the 21st century.
- Now that we can connect everyday objects—kitchen appliances, cars, thermostats, baby monitors—to the internet via embedded devices, seamless communication is possible between people, processes, and things.
- By means of low-cost computing, the cloud, big data, analytics, and mobile technologies, physical things can share and collect data with minimal human intervention.

Communication Technologies in IoT



Communication Technologies in IoT

- **Radio-Frequency Identification (RFID)**
  - **RFID** system is composed of one or more reader(s) and several RFID tags.
  - **RFID tags** are characterized by a specific address and applied upon objects.
    - They make use radio-frequency electromagnetic fields to transfer data associated to an object,
    - These tags are embedded with electronically stored information which can be read by RFID reader when the object came in the proximity of the reader.
  - **RFID** allows monitoring objects in real-time, without the need of being in line-of-sight.

Communication Technologies in IoT

- **Radio-Frequency Identification (RFID)**
  - An RFID tag comes in three configurations.
    - **Passive Reader Active Tag** (PRAT)
      - The reader is passive and receives the signal from the battery-operated active tags,
    - **Active Reader Passive Tag** (ARPT), which is most commonly used for communication purposes.
    - **Active Reader Active Tag** (ARAT).

Communication Technologies in IoT

- **IEEE 802.15.4**
  - It is a standard which specifies the physical layer and media access control for Low-Rate Wireless Personal Area Networks (LR-WPANs).
  - The original version of IEEE 802.15.4 supported 826 and 915 MHz frequency bands, while the working model supports 2.4 GHz ISM band.
  - The basic framework conceives a 10 m communications range with a transfer rate of 250 kbit/s

Communication Technologies in IoT

- **Zensys Wave (Z-Wave)**
  - Z-Wave is a low-power wireless communication protocol for Home Automation Networks (HAN).
  - It is used widely in remote control applications for smart homes, as well as small-size commercial domains.
  - In Europe, it operates at 868.42 MHz, whereas in Canada and the USA it uses 908.42 MHz bandwidth for communication.
  - The distance between two nodes should not be exceeding 30 meters.

Communication Technologies in IoT

- **Long Term Evolution (LTE)**
  - **LTE** is a standard wireless communication protocol for high-speed data transfer between mobile phones based on Global System for Mobile Communication (GSM) network technologies.
  - It supports up to a maximum of 100 MHz
  - Data upload and download often encounter lower latency rate and higher throughout.

Communication Technologies in IoT

- **Near-Field Communication (NFC)**
  - **NFC** is quite similar to RFID.
  - It is overlooked as the integration of RFID readers onto a mobile phone.
  - **NFC** is also be seen as a unique kind of a radio communication device that is enabled on mobile devices either by tracing inbuilt options that need to be activated or by them two devices close in the proximity.
  - **NFC** is a short-range, low power wireless link evolved from RFID that can transfer small amounts of data between two devices held in proximity.
  - It enables the communication between two smart objects, that is safe, as this cannot be done from a remote location.

Communication Technologies in IoT

- **Ultra-wide Band (UWB)**
  - **UWB** communication technology is designed to support communications within low range coverage areas, that are similar to NFC that uses low energy.
  - However, high bandwidth is used for applications to connect sensors for communication.
  - It is capable of a maximum of 500 MHz bandwidth. It was earlier known as radio pulse.

Communication Technologies in IoT

- **Machine to Machine (M2M)**
  - M2M refers to the communications either between computers, embedded processors, smart sensors, actuators or with mobile devices.
  - There are a total of five basic components for any M2M communication technique,
    - i.e., sensing, heterogeneous access, information processing, applications, and services.

Communication Technologies in IoT

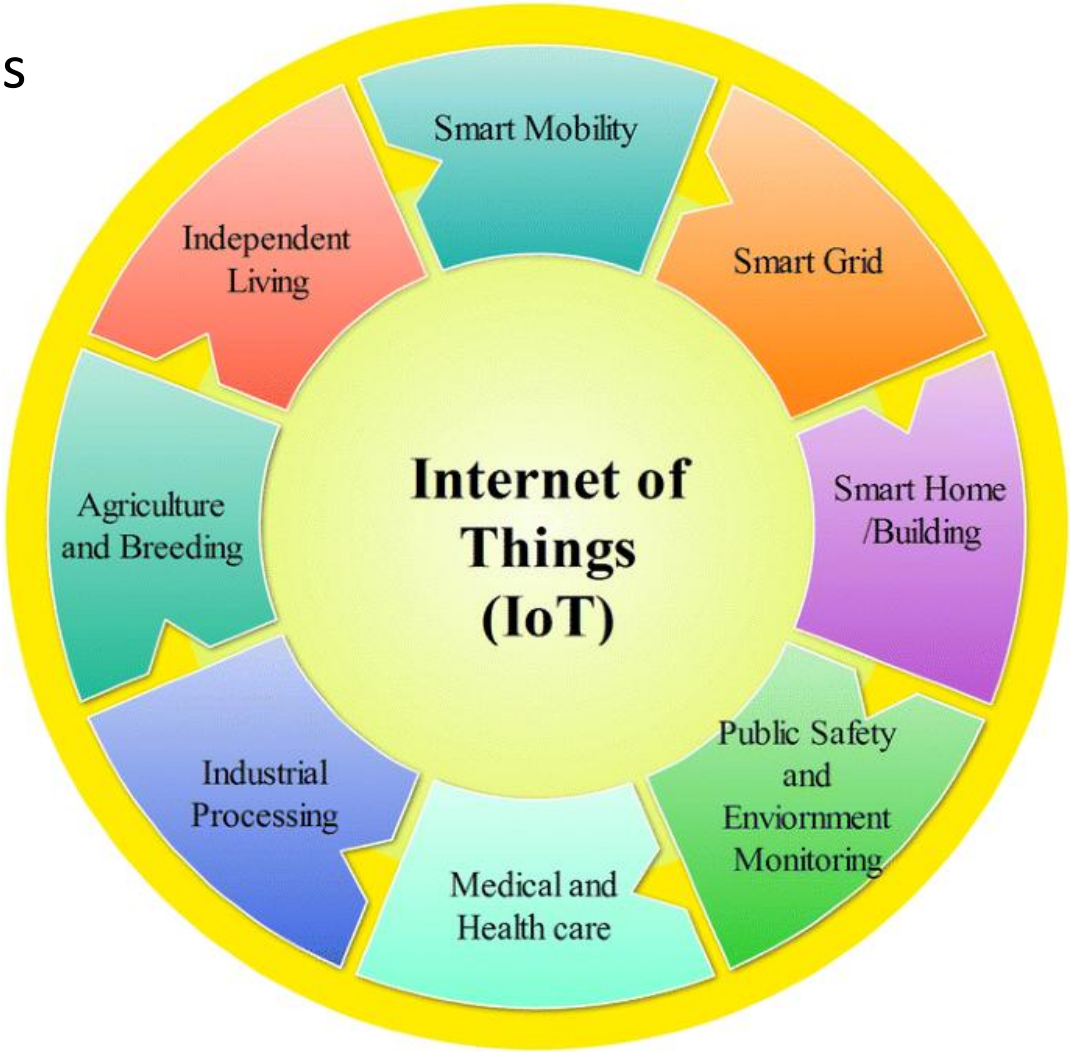
- **Machine to Machine (M2M)**
  - M2M is a five-part structure that constitutes of following parts:
    1. **M2M Device**: A device capable of replying to request for data contained within that device.
    2. **M2M Area Network** (Device Domain): Provide connectivity between M2M Devices and M2M Gateways.
    3. **M2M Gateway**: Use M2M capabilities to ensure M2M Devices inter-working and interconnection to the communication network.
    4. **M2M Communication Networks** (Network Domain): Communications between the M2M Gateway(s) and M2M application.
    5. **M2M Applications**: Contains the middleware layer where data goes through various application services and is used by the specific business-processing engines.

Communication Technologies in IoT

- **IPv6 Low-Power Wireless Personal Area Network (6LoWPAN)**
  - 6LoWPAN enables IPv6 to run on low-power wireless networks.
  - This network protocol defines encapsulation and header compression mechanisms that allow IPv6 to operate efficiently on resource-constrained devices.
  - The standard is designed to be independent of the physical layer and can operate across multiple communication platforms,
    - including IEEE 802.15.4, Bluetooth Low Energy (BLE), ITU-T G.9959 (Z-Wave), and DECT-ULE.

IoT Apps

- Smart Thermostats
- Connected Cars
- Activity Trackers
- Parking Sensors
- Connect Health
- Smart City



Challenges of IoT

- Concern regarding data security and privacy.
- IoT Software complexity: Insufficient testing and updating.
- Data volumes and interpretation.
- Devices require a constant power supply which is difficult.

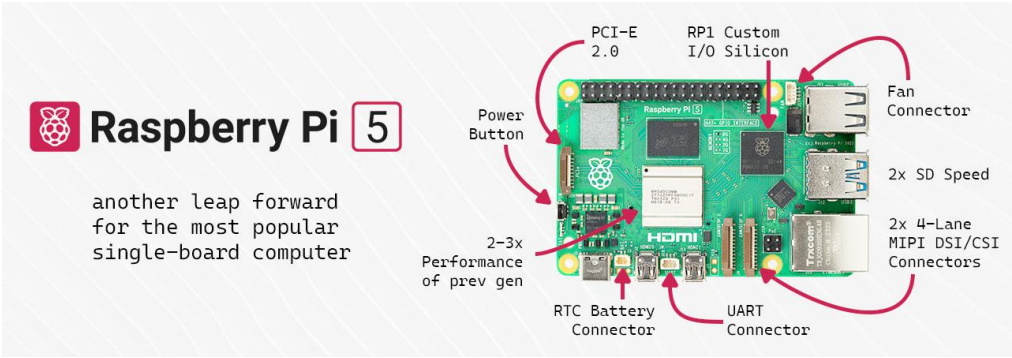
Challenges of IoT

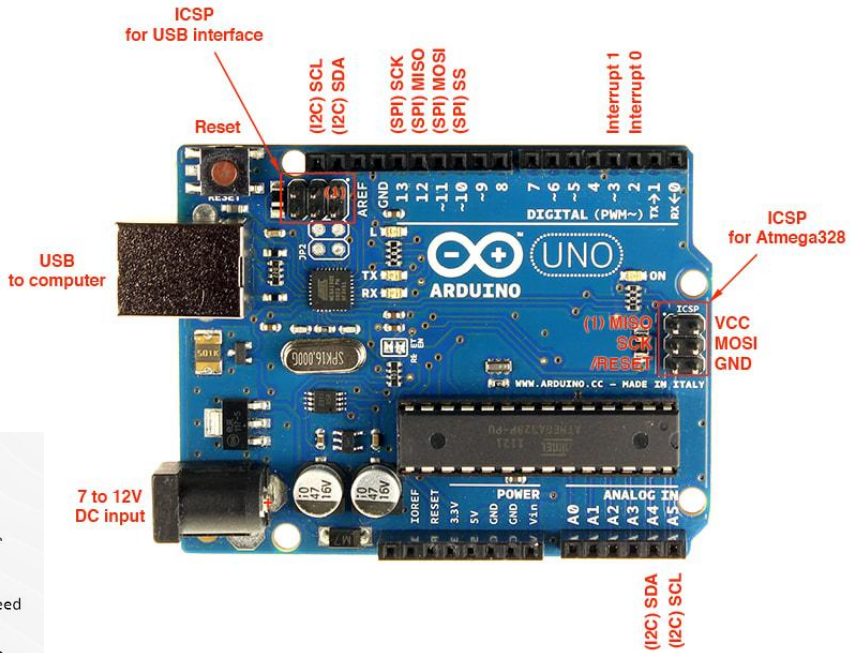
- **Data collection and management!!**

  IoT devices generate massive amounts of complex data, so they must be secured against breaches and protected under data privacy laws.

Microcontrollers To Design IoT

- Arduino
- Raspberry Pi
- ESP8266 and ESP32





Arduino

Future for IoT

1. By 2025, it is estimated that there will be more than to 21 billion IoT devices
2. Cybercriminals will continue to use IoT devices to facilitate DDoS attacks
3. More cities will become “smart”
4. Routers will continue to become more secure and smarter
5. 5G Networks will continue to fuel IoT growth
6. Security and privacy concerns will drive legislation and regulatory activity

References

- Khanna, Abhishek, and Sanmeet Kaur. "Internet of things (IoT), applications and challenges: a comprehensive review." Wireless Personal Communications 114 (2020): 1687-1762.